

Correlation and Covariance using `cor_cov`

Siddhesh Akole

February 16, 2026

This vignette demonstrates the `cor_cov` function from the **numTheory** package. The function computes both correlation and covariance between two numeric vectors. Correlation measures the strength of linear association, while covariance measures the joint variability of two variables.

1 Inputs

The function takes two numeric vectors, `x` and `y`, as input.

- `x` and `y` are the data vectors whose relationship is being analyzed.
- The output includes both correlation and covariance values.

2 Basic Example

Here we compute correlation and covariance for two simple vectors:

```
> library(numTheory)
> x <- c(1,2,3,4,5)
> y <- c(2,4,6,8,10)
> # Correlation and covariance of x and y
> cor_cov(x, y)

$covariance
[1] 5

$correlation
[1] 1
```

3 Interpreting Results

In this example, the correlation is 1 because `y` is exactly proportional to `x`. The covariance is positive, reflecting that both variables increase together.

```
> # Perfect proportionality: y = 2x
> x <- c(1,2,3,4,5)
> y <- 2 * x
> cor_cov(x, y)
```

```
$covariance  
[1] 5
```

```
$correlation  
[1] 1
```

4 Edge Cases

The function also handles cases where vectors have zero variance or mismatched lengths.

```
> # Zero variance in y  
> y2 <- c(5,5,5,5,5)  
> cor_cov(x, y2)
```

```
$covariance  
[1] 0
```

```
$correlation  
[1] NaN
```

```
> # Mismatched lengths (expected: error)  
> y3 <- c(1,2,3)  
> try(cor_cov(x, y3))
```

5 Summary

The `cor_cov` function provides a convenient way to compute correlation and covariance together, making it easy to analyze relationships between variables.